

TAKFA AGROMET, Thailand

WMO: 484010

Lat: 15.3498N

Lon: 100.5303E

Elev: 308

StdP: 14.53

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	63.3	66.1	48.3	50.6	70.1	51.5	57.2	72.6	9.8	74.4	9.2	75.7	1.5	N/A	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.9	100.6	78.5	98.7	78.9	96.9	78.7	81.9	92.4	81.5	91.7	81.0	90.8	3.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.7	156.1	86.0	79.1	152.9	85.5	78.7	150.7	85.2	46.1	93.0	45.5	91.7	45.0	91.0	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.7	7.5	6.8	DB	57.9	102.1	4.1	2.7	55.0	104.0	52.6	105.6	50.3	107.1	47.4	109.1	(4)
			WB	53.6	83.6	4.3	1.0	50.5	84.3	48.0	84.9	45.6	85.5	42.4	86.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	79.0	82.9	86.1	88.2	87.1	85.3	84.0	83.5	83.0	82.8	81.8	78.9
	DBStd	4.03	4.06	3.13	3.76	3.44	3.18	2.18	2.20	1.61	1.76	1.93	2.63	3.97
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	1	1	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12241	898	921	1119	1145	1149	1060	1053	1039	991	1016	953	897
	CDD65	6767	434	501	654	695	684	610	588	574	541	551	503	432
	CDH74	77613	4183	5700	8478	9463	8886	7391	6643	6176	5699	5767	5195	4032
	CDH80	35384	1684	2693	4497	5418	4668	3394	2772	2411	2093	2159	2058	1537
Wind	WSAvg	1.6	1.9	1.6	1.8	1.9	1.5	1.6	1.4	1.1	0.9	1.3	2.2	2.6
Precipitation	PrecAvg	45.7	0.3	0.4	1.7	2.9	5.9	5.2	5.4	6.9	10.0	5.7	1.0	0.2
	PrecMax	59.3	1.4	2.1	5.4	6.5	10.2	7.8	9.6	11.4	15.0	16.3	4.1	1.8
	PrecMin	31.7	0.0	0.0	0.0	0.4	1.3	1.6	2.3	3.5	6.1	0.3	0.0	0.0
	PrecStd	6.6	0.4	0.5	1.3	1.7	2.2	1.8	1.7	1.7	2.4	3.0	1.1	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	94.0	97.0	100.6	103.8	102.4	98.8	97.9	94.1	93.4	93.7	93.5	94.0
		MCWB	74.2	76.6	75.8	77.9	80.3	79.8	79.0	80.1	80.1	78.7	77.9	75.0
	2%	DB	91.9	95.1	98.6	101.7	99.6	96.0	94.0	92.3	92.0	92.0	92.2	91.9
		MCWB	73.9	76.1	76.9	78.4	79.9	79.7	79.2	79.7	79.9	78.5	77.4	75.4
	5%	DB	90.3	93.4	96.8	99.3	97.6	93.9	92.0	90.9	90.6	90.6	90.9	90.1
		MCWB	73.1	75.6	77.2	79.1	79.9	79.8	79.2	79.4	79.8	78.3	76.7	74.7
	10%	DB	88.5	91.7	94.9	97.2	95.1	91.9	90.2	89.5	89.0	89.2	89.3	87.9
		MCWB	72.5	75.2	77.4	79.2	80.0	79.6	79.1	79.3	79.6	78.3	75.9	73.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.9	80.5	81.3	82.7	83.0	82.0	81.7	81.8	81.9	81.6	81.0	79.2
		MCDB	88.0	93.5	93.9	95.5	94.4	92.1	90.2	89.9	89.5	89.1	90.1	88.9
	2%	WB	77.4	79.1	80.2	81.7	82.0	81.2	80.8	81.0	81.2	80.8	80.1	77.8
		MCDB	86.4	90.7	92.3	94.5	93.0	91.1	89.5	89.0	88.7	88.2	88.5	87.4
	5%	WB	76.4	78.2	79.5	81.0	81.3	80.7	80.2	80.5	80.7	80.1	79.0	76.4
		MCDB	85.2	88.9	91.2	93.4	92.0	90.4	88.6	88.3	88.2	87.1	86.6	85.9
	10%	WB	75.2	77.3	78.8	80.4	80.7	80.1	79.7	79.9	80.0	79.4	77.9	75.1
		MCDB	83.6	87.1	90.2	92.4	91.2	89.1	87.8	87.3	87.0	86.0	85.1	84.8
Mean Daily Temperature Range	5% DB	MDBR	18.4	18.6	17.9	17.9	15.5	14.0	13.2	13.1	12.4	13.1	15.0	16.9
		MCDBR	21.0	20.4	20.3	20.1	18.4	16.5	15.6	14.8	14.4	15.1	16.9	18.7
		MCWBR	7.9	6.9	5.6	5.3	4.6	4.5	4.5	4.8	5.0	4.9	5.9	6.8
	5% WB	MCDBR	16.3	17.8	18.1	18.2	16.5	14.7	13.8	13.9	13.4	13.6	15.2	17.0
	MCWBR	6.2	6.1	5.5	5.5	5.3	4.9	4.8	5.2	5.2	5.0	5.6	6.9	
Clear-Sky Solar Irradiance	taub	0.510	0.599	0.663	0.598	0.500	0.448	0.438	0.442	0.480	0.553	0.492	0.492	
	taud	2.004	1.812	1.702	1.890	2.213	2.367	2.392	2.365	2.245	1.994	2.139	2.072	
	Ebn at Noon	247	231	221	236	256	267	271	272	263	240	251	249	
	Edh at Noon	54	69	78	65	46	39	38	40	45	56	47	49	
All-Sky Solar Radiation	RadAvg	1531	1671	1747	1899	1816	1720	1630	1588	1532	1555	1559	1508	
	RadStd	94	63	105	79	104	90	79	94	74	85	74	53	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	+0.41	+0.47	N/A	N/A	+180	+176

Nomenclature: See separate page